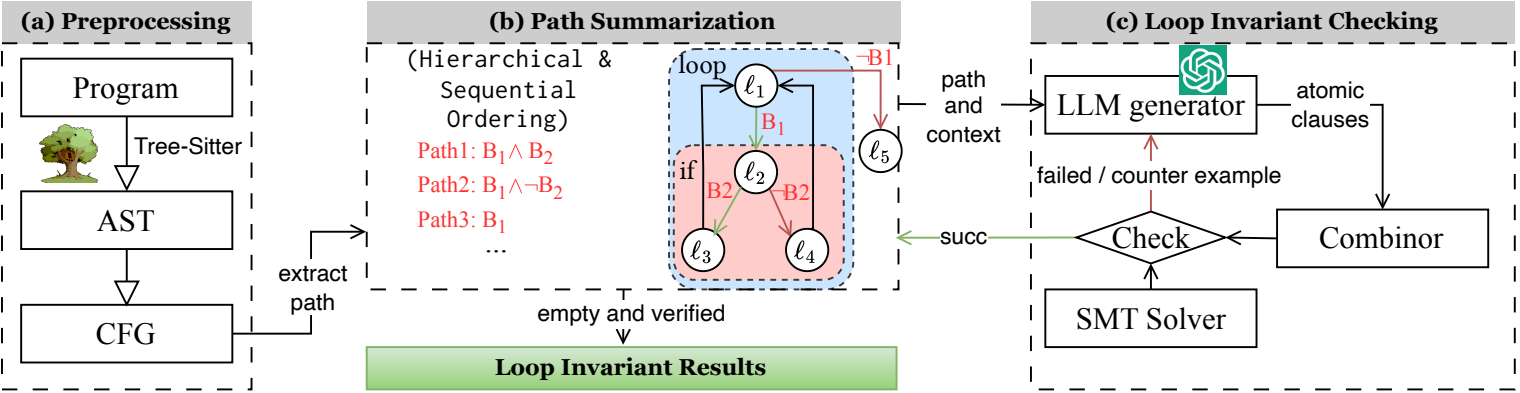
(a) Preprocessing
Program
Tree-Sitter
AST
CFG
extract path
(b) Path Summarization
(Hierarchical & Sequential Ordering)
Path1: B1∧B2
Path2: B1∧¬B2
Path3: B1
...
loop
if
¬B1
B1
B2
¬B2
empty and verified
Loop Invariant Results
path and context
succ
(c) Loop Invariant Checking
LLM generator
atomic clauses
failed / counter example
Check
Combinor
SMT Solver